Examiner only

**2.** *(a)* A battery does not *store* charge. State what a battery *does* do in relation to charge in an electric circuit. [1]

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*(b)* A battery consists of three cells, **each** of emf 1.60 V and internal resistance 0.10 Ω, connected in series. The battery is connected to an electromagnet of resistance 1.20 Ω.

(i) Show clearly that the current is approximately 3 A. [The space is for a diagram if required.] [3]

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(ii) Calculate the rate (in watts) at which:

I. energy is dissipated by the electromagnet; [1]

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II. the battery's chemical energy is being used. [1]

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(iii) The answer to *(b)*(ii)II. is expected to be greater than the answer to *(b)*(ii)I. Explain where the missing energy goes. [2]

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Examiner only

(c) A student wishes to maximise the current through the electromagnet. She has a spare cell of emf 1.50 V and internal resistance 0.50 Ω. Evaluate whether or not she should put it in series with the battery, giving your calculations and conclusion clearly. [2]

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